

Reading Assignment

1. What are the advantages of Polymorphism?

Polymorphism allows objects to behave in multiple ways, even if they share the same interface. This brings several benefits:

- **Code reusability:** The same method can work with different types of objects, reducing duplication.
- **Flexibility and scalability:** We can easily add new classes with specific behavior without changing existing code.
- **Improved maintainability:** Changes in one class do not affect other parts of the system that use the common interface.
- **Runtime decision making:** Through method overriding, the program decides which version of a method to run during execution (dynamic polymorphism).

2. How is Inheritance useful to achieve Polymorphism in Java?

Inheritance is the foundation for polymorphism in Java. When a subclass inherits from a superclass, it gains all non-private members. This enables:

- **IS-A relationship:** A subclass object can be treated as an instance of its superclass.
- **Method overriding:** A subclass can provide a specific implementation of a method defined in the superclass. When using

superclass references, the overridden method in the subclass is called at runtime.

This behavior allows objects to take on different forms and still interact through a shared interface.

3. What are the differences between Polymorphism and Inheritance in Java?

Aspect	Inheritance	Polymorphism
Definition	Mechanism where one class inherits from another	Ability of an object to take many forms
Purpose	Code reuse and class hierarchy	Dynamic behavior and interface sharing
Focus	Structure (parent-child relationship)	Behavior (method overriding and overloading)
Dependency	Polymorphism often relies on inheritance	Can exist without inheritance in the case of overloading
Example feature	“extends” keyword in class declaration	Method overriding (runtime), method overloading (compile-time)